

Shanilka Haturusinghe

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EDUCATION

University of Kelaniya, Sri Lanka

March 2020 – October 2024

Bachelor of Science Honours in Computer Science (*First Class*)

Specialization: Artificial Intelligence

GPA: 4.0/4.0

• Graduated as the **top-ranked student (1st overall)** in the Faculty of Computing and Technology's Class of 2024, outperforming all peers across the faculty's three undergraduate degree programs.

• **Recipient of full-tuition Merit Scholarship from the Government of Sri Lanka**, awarded to only 447 students island-wide for B.Sc. Computer Science programs based on performance in the 2018 national university entrance examination [GCE A-Levels] (32,304 candidates in the Physical Science stream).

PUBLICATIONS

Shanilka Haturusinghe, Tharindu Cyril Weerasooriya, Christopher M Homan, Marcos Zampieri, and Sidath Ravindra Liyanage. 2025. *Subasa – Adapting Language Models for Low-resourced Offensive Language Detection in Sinhala*. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 4: Student Research Workshop)*, pages 260–270, Albuquerque, USA. Association for Computational Linguistics. DOI: 10.18653/v1/2025.naacl-srw.26

Link: <https://aclanthology.org/2025.naacl-srw.26/>

- Developed Subasa-XLM-R model by adapting Masked Rationale Prediction(MRP) as an intermediate pre-finetuning task on xlm-roberta-base, then fine-tuned for binary classification, achieving 0.84 Macro F1 on SOLD(Sinhala Offensive Language Detection) benchmark.
- Developed Subasa-Llama variants (3.2-3B and 3.1-8B) and Subasa-Mistral-7B via task-specific(joint offensive classification and phrase extraction from rationale spans) fine-tuning with PEFT, outperforming base models and GPT-4o zero-shot by up to 24% Macro F1.
- Performance on SOLD benchmark: Subasa-XLM-R at 0.84 Macro F1, Subasa-Llama-3.1-8B at 0.826, surpassing GPT-4o (0.733) and multilingual baselines like Aya101, (effectiveness of method verified with rigorous ablation studies).
- Publicly released all models, code. Access via <https://github.com/haturusinghe/subasa-llm> and <https://github.com/haturusinghe/subasa-plm>
- Key Technologies: PyTorch, Hugging Face Transformers and Datasets, Unsloth, Google Cloud Platform

Ushara Prabash Christy Melder, **Shanilka Haturusinghe**, and S.P. Kasthuri Arachchi. *Customer segmentation and churn prediction in the maritime shipping industry using machine learning techniques: A Sri Lankan case study*. *Currently Under Review, Submitted November 2025* ; Link to Manuscript: <http://bit.ly/48oE3CR>

- A machine learning framework for customer churn prediction and unsupervised customer segmentation, developed and validated using a newly created real-world dataset compiled from Sri Lanka maritime shipping company data.
- Key results: XGBoost outperforms other models on the Sri Lankan dataset, achieving an accuracy of 0.854, churn recall of 0.94, F1-score of 0.89, and AUC-ROC of 0.89, compared to Random Forest (accuracy 0.851, AUC-ROC 0.88) and Logistic Regression (accuracy 0.70, AUC-ROC 0.69).
- Streamlit-based front-end for generating churn predictions and integrated with Power BI for real-time visualization of churn probabilities and customer segments.
- Key Technologies: Python, Scikit-learn, Pandas, XGBoost, Random Forest, Logistic Regression, Streamlit

TEACHING EXPERIENCE

University of Kelaniya

August 2025 - Present

Lecturer

- Lecturer-in-Charge for CSCI 31032 - **Theory of Programming Languages**: responsible for delivering all lectures, office hours, assignments, grading, and examinations for this module.
- Contributing lecturer for CSCI 31022 - **Machine Learning and Pattern Recognition**; Contributing lecturer for CSCI 11532 - **Structured Programming I**.

University of Kelaniya

February 2025 - July 2025

Teaching Assistant & Demonstrator

- Served as TA for: *DSCI 32012 - Advanced Database Applications (70 students)*, *CTEC 22043 - Object Oriented Programming (50 students)*, *AINT 32012 - Digital Image Processing and Computer Vision (10 students)*, *CSCI 22082 - Object-Oriented Programming II (50 students)*. Conducted office hours, led labs, and developed and graded assignments.

INDUSTRY EXPERIENCE

Embla Software Innovation (Pvt) Ltd

October 2023 - April 2024

Software Engineer Intern

- R&D contributions to Embla FT AI Helpdesk feature by prototyping and evaluating a RAG-based helpdesk system. Worked across multiple sprints gaining practical experience with LLMops and building RAG systems.
- Developed and implemented the "Scrapbook" feature for Embla FT genealogy application with customizable page templates, editable text/photo elements, and comprehensive image/content management capabilities.
- Built full-stack components using C#/.NET Core backend and Angular 9/TypeScript frontend, enhancing existing application architecture to support new scrapbook functionality and user interactions.
- Collaborated with product teams using Agile methodologies, Jira, and Bitbucket for project delivery, utilizing Visual Studio, FastReport Designer, and Figma for development and design tasks.

Xempler Pte. Ltd

September 2022 - May 2023

Intern - Software Developer

- Developed a Sri Lanka License Plate Detection proof of concept using Python and TensorFlow by leveraging YOLO models for robust license plate detection in images. Employed OpenCV for image preprocessing (resizing, filtering, and edge detection) to optimize detection accuracy, and integrated Tesseract OCR to extract alphanumeric text from the detected license plates.

	Designed and implemented a real-time chat feature for the IFAWPCA mobile app, built with React Native. Integrated Firebase for seamless real-time messaging, utilizing Firebase Realtime Database and Firebase Authentication for user management, secure login, and instant message synchronization across devices.
AWARDS AND HONORS	Gold Medal for Academic Excellence (2025) – Awarded by the Alumni Association of the University of Kelaniya. Gold Medal for Outstanding Performance (2025) – Awarded by the Faculty of Computing and Technology. Google Cloud Research Credits Program (Award GCP19980904) – Awarded \$1000 in research credits - First Class Honors - BSc in Computer Science - Distinction in Combined (Pure & Applied) Mathematics, GCE A-Levels (2018) - in top 8.58% nationwide (out of 35,678 candidates) for Combined Mathematics.
PERSONAL PROJECTS	PersonaCraft.AI - Personalized AI Writing Assistant https://github.com/haturusinghe/persona-craft-ai - Developed an end-to-end system for creating personalized AI writing assistants through automated data collection, processing, and model fine-tuning with multi-platform web scraping capabilities. - Built automated data extraction pipelines using Selenium for LinkedIn, Medium, GitHub, and custom web sources, implementing advanced text processing with chunking and embedding generation via Sentence Transformers. - Engineered training pipeline supporting both Supervised Fine-Tuning (SFT) and Direct Preference Optimization (DPO) using TRL and Unsloth,. - Implemented MLOps architecture using ZenML for pipeline orchestration, MongoDB for data warehousing, Qdrant vector database for RAG-based retrieval, and AWS SageMaker for cloud-based model training. Key Technologies: Python, ZenML, LangChain, GPT-4, Llama 3.1, TRL, Unsloth, MongoDB, Qdrant, AWS SageMaker, Selenium, Docker, Poetry, Comet ML Real-Time Fashion Recommendation System https://github.com/haturusinghe/realtime-fashion-recommender - Developed a MLOps-driven fashion recommendation system using H&M's Kaggle dataset, implementing a 4-stage architecture with decoupled ML pipelines for feature processing, training, and inference. - Implemented a two-tower neural network architecture for customer&item embedding generation, utilizing Hopworks Feature Store for embedding storage and retrieval. - Built comprehensive ML pipelines including feature engineering, retrieval model training (two-tower architecture), ranking model training (CatBoost), item embedding computation, and model deployment with real-time inference capabilities. - Integrated Hopworks Feature Store for feature management, Model Registry for versioning, and Model Serving for deployment, with an interactive Streamlit frontend for customer recommendations and feedback. Key Technologies: Python, TensorFlow, Hopworks, Sentence-Transformers, CatBoost, Polars, Streamlit Deep Learning based real-time Facial Age and Gender Detection System https://github.com/haturusinghe/cnn-age-gender-detection-android - Developed CNN models for age and gender detection, integrated into Android for real-time predictions. Backend implemented with TF-Serving and Flask. - Deployed system using TensorFlow for model inference and integrated a model-serving pipeline with preprocessing. Ensured real-time interaction through mobile interfaces. Key Technologies: Python, Flask, TensorFlow (+ TF-Lite, TF-Serving), Android Studio, Kotlin, Heroku
OTHER ACHIEVEMENTS	Huawei Seeds for the Future - 2023 - Team Leader for Team Sri Lanka at Huawei Seeds for the Future - 2023 – Selected as one of only four top university students from a highly competitive pool, earning the opportunity to visit China. - Participated in Training program, received mentorship from top industry experts on cutting-edge technologies including 5G, AI, Cloud Computing, and Digital Power.
ENGLISH PROFICIENCY	IELTS Academic - 2025 Overall Band Score: 8.5 - Listening: 9.0 Reading: 8.5 Writing: 7.5 Speaking: 8.0
REFERENCES	Prof. S. R. Liyanage Professor, Department of Software Engineering, Faculty of Computing and Technology, University of Kelaniya Email: sidath@kln.ac.lk, Phone: +94 71 649 8614 Dr. Tharindu Cyril Weerasooriya Research Scientist, Accenture (USA) Work Email: t.weerasooriya@accenture.com, Personal Email: cyrilcw@gmail.com , Phone: +1 585 967 6292 Website: https://cyrilw.com/ Dr. S. P. Kasthuri Arachchi Head of the Department of Software Engineering , Faculty of Computing and Technology, University of Kelaniya Email: sandelik@kln.ac.lk, Phone: +94 74 293 5784